

Table 8: *Chain-of-thought (CoT) prompt templates*. One of the six chain-of-thought prompt templates we use for the CoT experiment. Note that this is a 5-shot prompt. Each prompt variation contains five CoT examples. The other five variations are separately added to the supplementary materials

#	Prompt template
1	Bob asks Alice a question, and Alice responds with an implicature. This means that Alice’s response does not literally contain the answer to Bob’s question, but implies an answer. Assuming that Alice is a cooperative conversational partner, what is the implicated answer to the question? For example: Bob: You invented fire? Alice: I told you that. Implicature: Alice says ‘I told you that’. Alice’s response must be relevant to Bob’s question because Alice is a cooperative conversational partner. Alice must mean that she told Bob that she invented fire. Alice’s response to Bob’s question ‘You invented fire?’ is yes. Answer: yes Bob: That cake looks delicious. Aren’t you going to have some with me? Alice: But that was a huge meal we just had. Implicature: Alice’s response must be relevant to Bob’s question because Alice is a cooperative conversational partner. Alice must mean that the meal they just had was so huge she is not hungry anymore, and this must be relevant to Bob’s question: ‘Aren’t you going to have some with me?’ Alice’s response to the question must therefore be no. Answer: no Bob: Could you perform well? Alice: Being bilingual would help put me ahead of the pack. Implicature: Alice says being bilingual would help put her ahead of the pack. Alice’s response must be relevant to Bob’s question because Alice is a cooperative conversational partner. Alice must be implying that she could perform well because she is bilingual. This means the response to Bob’s question is yes. Answer: yes Bob: Have you any news for me? Alice: I’ve made progress Implicature: Alice says she has made progress. Alice’s response must be relevant to Bob’s question because Alice is a cooperative conversational partner. If Alice would not have any news for Bob, Alice would not have said she would have made progress because that would be misleading. The answer to Bob’s question ‘Have you any news for me?’ must therefore be yes. Answer: yes Bob: You looked out for him? Alice: He looked out for me. He taught me. Implicature: Bob asks Alice ‘You looked out for him?’ and Alice’s response says that the person that is being referred to by ‘him’ here looked out for Alice. If Alice meant yes to Bob’s question, Alice would have said something like ‘he also looked out for me’. Stating the response like this implies that the answer to Bob’s question is no. Answer: no Only output a ‘yes’ or ‘no’ as a final answer. Write your reasoning after ‘Implicature:’ and then output either ‘Answer: yes’ or ‘Answer: no’. Bob: <utterance> Alice: <response> Implicature:

G Model categorization

Table 9 contains details on the model classes that are a part of each group of models we evaluate, along with their model sizes.

Table 9: Model categorization for each of the models. DL stands for dialogue, FT for fine-tuning, BL for benchmark-level, EL for example-level, and IT for instruction-tuning.

Group	Model class	Model IDs	Model size	Instruct
Base	BERT	base uncased	110M	No
	RoBERTa	base, large	125M, 355M	No
	GPT-2	GPT-2 medium, large, xl	354M, 774M, 1.6B	No
	EleutherAI	GPT-J, GPT-NeoX	125M, 1.3B, 2.7B, 6B, 20B	No
	BLOOM	-	560M, 1B1, 3B, 7B1, 176B	No
	OPT	-	125M, 350M, 1.3B, 13B, 30B, 66B, 175B	No
	Cohere	small, medium, large, XL	409.3M, 6.067B, 13.12B, 52.4B	No
	GPT-3	ada, babbage, curie, davinci	Est. 350M, 1.3B, 6.7B, 175B	No
DL FT	BlenderBot	-	90M, 2.7B, 9.4B	No
BL IT	T0	-	3B, 11B	Yes
	Flan-T5	-	780M, 3B, 11B	Yes
EL IT	Cohere-command	medium, xlarge	6.067B, 52.4B	Yes
	text-davinci-001	ada, babbage, curie, davinci-1	Unknown, left-to-right increasing in size	Yes
	text-davinci-002	-	Unknown	Yes
	text-davinci-003	-	Unknown	Yes
	ChatGPT	gpt-3.5.turbo	Unknown	Yes
	GPT-4	gpt-4	Unknown	Yes

H Human evaluation

The participants for the human evaluation in this paper were recruited using Prolific (www.prolific.co). The setup of the experiment is as follows. We divide the test set of 600 examples into four non-overlapping subsets of 150 examples. Each set of 150 examples was given to five unique annotators. This means each example in the test set is labeled five times by different people, and we have in total twenty annotators for the whole test set (five different ones for each of the four subsets). The only constraint for the annotators is that they are native English speakers. In Figure 5 the screen shown to potential participants on Prolific is shown. Participants are paid 15 pounds an hour, which was the living wage at the time of the experiment and more than the 12 dollars an hour Prolific recommends. The total amount spent on the human evaluation is 236 pounds. This amount came to be from four subsets, each costing about 30 minutes to label per annotators, and having 5 annotators per subset: $15 * 4 * 0.5 * 5 = 150$. The extra costs were for the annotator that didn't pass the attention check which we paid nonetheless, and for prolific as a platform.

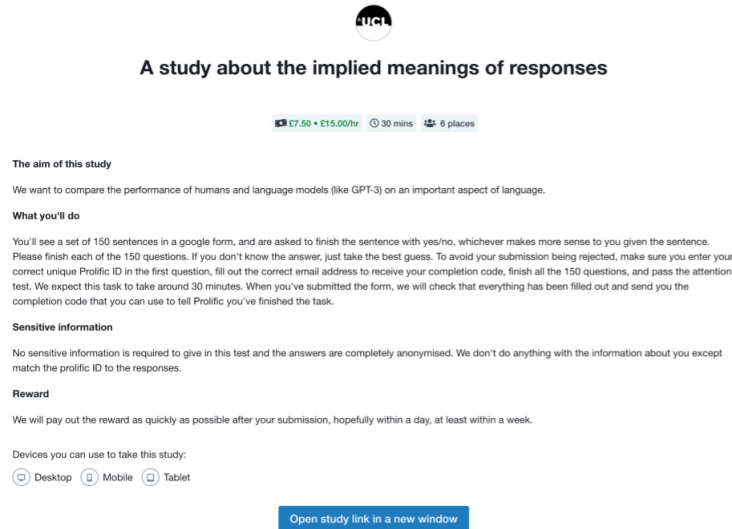


Figure 5: A screenshot of how the experiment is presented to potential annotators on Prolific (www.prolific.co).

The 150 test examples are wrapped in prompt template 2 (see Table 5) and presented in a Google form. We opted to wrap all examples in prompt template 2 to make the full human study directly comparable to the model's results on template 2. If we had done a mix of all templates we either had to spent six times as much on the human evaluations (which was not within our budget) or subsample evaluations, making it less comparable to part of the model study. Although models have been shown to be very sensitive to prompt wording, humans are less likely to perform differently for different prompt templates. All templates are coherent natural language that any native English speaker will understand. That said, this is speculative, and to confirm this hypothesis future work should investigate the effect of different wordings on implicature resolution by humans. The participants are asked to choose the correct continuation, yes or no (see Figure 6a). As recommended by Prolific, we subject the participants to an attention test (see Figure 6b). At three random places in the form, we add a question that does not contain an implicature and obviously maps to "yes". In this way, if the participants fails at least two of these questions, we can conclude they were not paying attention and remove their answers from the result. In practice, this happened once and we decided to pay the participant regardless, but discard their results, which were close to random.

Table 10 shows the performance of each annotator on the subset they annotated. The average human performance across subsets and annotators is $86.2\% \pm 2.3$, the best performance is $89.8\% \pm 2.2$, and the worst performance is $83.5\% \pm 1.5$. The column "IAA" shows the average Cohen's Kappa coefficient which is the pairwise inter-annotator agreement for each annotator per subset. All agreements are substantial according to the interpretation guidelines for Cohen's Kappa (between 0.61–0.80).